



Current Regulations Driving MLOps Governance

A Strategic Framework for Building Trustworthy
and Scalable AI

Arindam Gupta

1RV23AI021

Ashish R Biradar

1RV23AI022

Kavya Jain

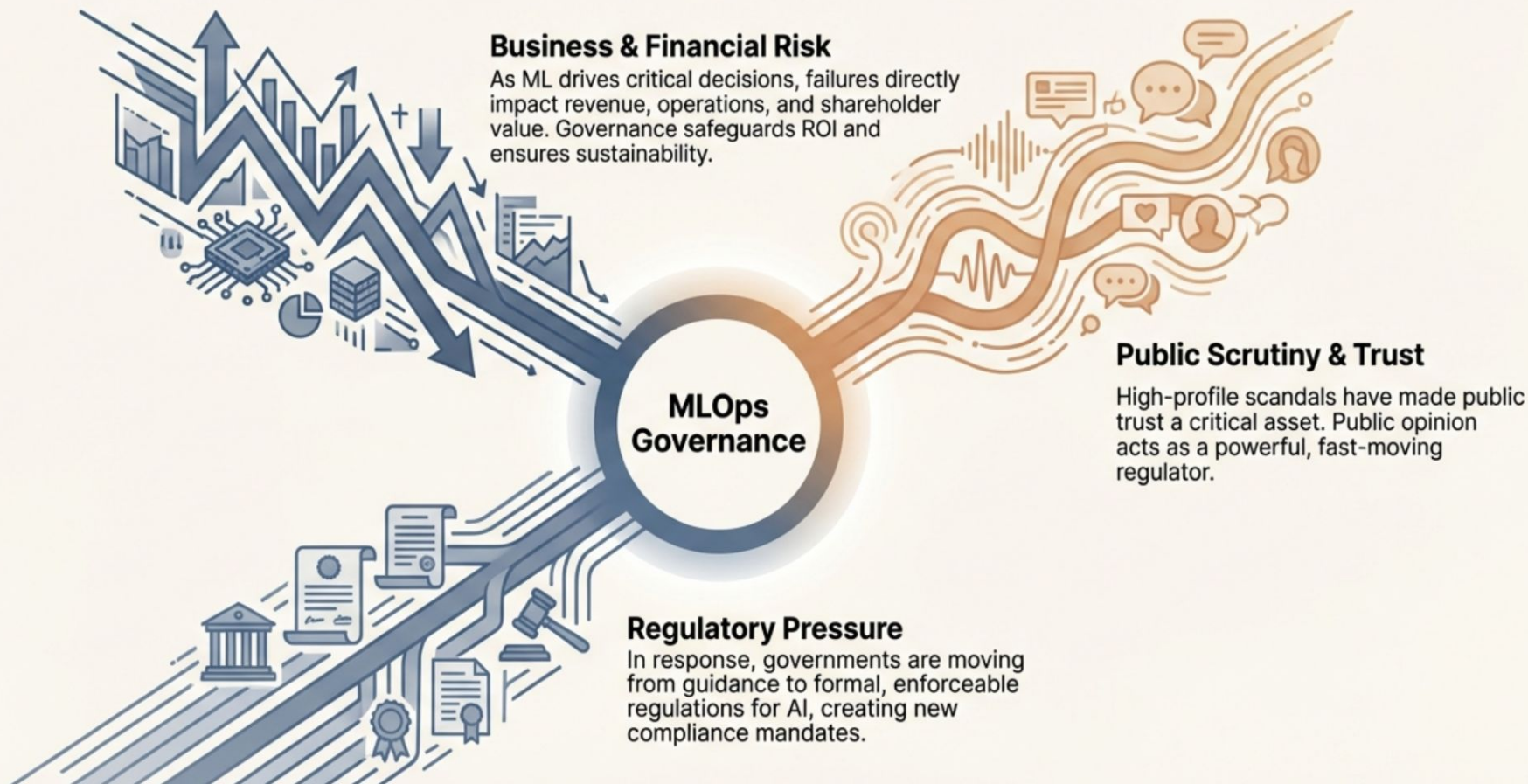
1RV23AI045

N Mohammed Akhil

1RV23AI061

Department of AIML

AI at Scale Creates a Perfect Storm of Risk



Deconstructing the Three Layers of Regulation Impacting MLOps

The regulatory landscape isn't a monolithic entity; it's a multi-layered structure built over time. Understanding these layers is the key to a comprehensive strategy.

The AI-Centric Future

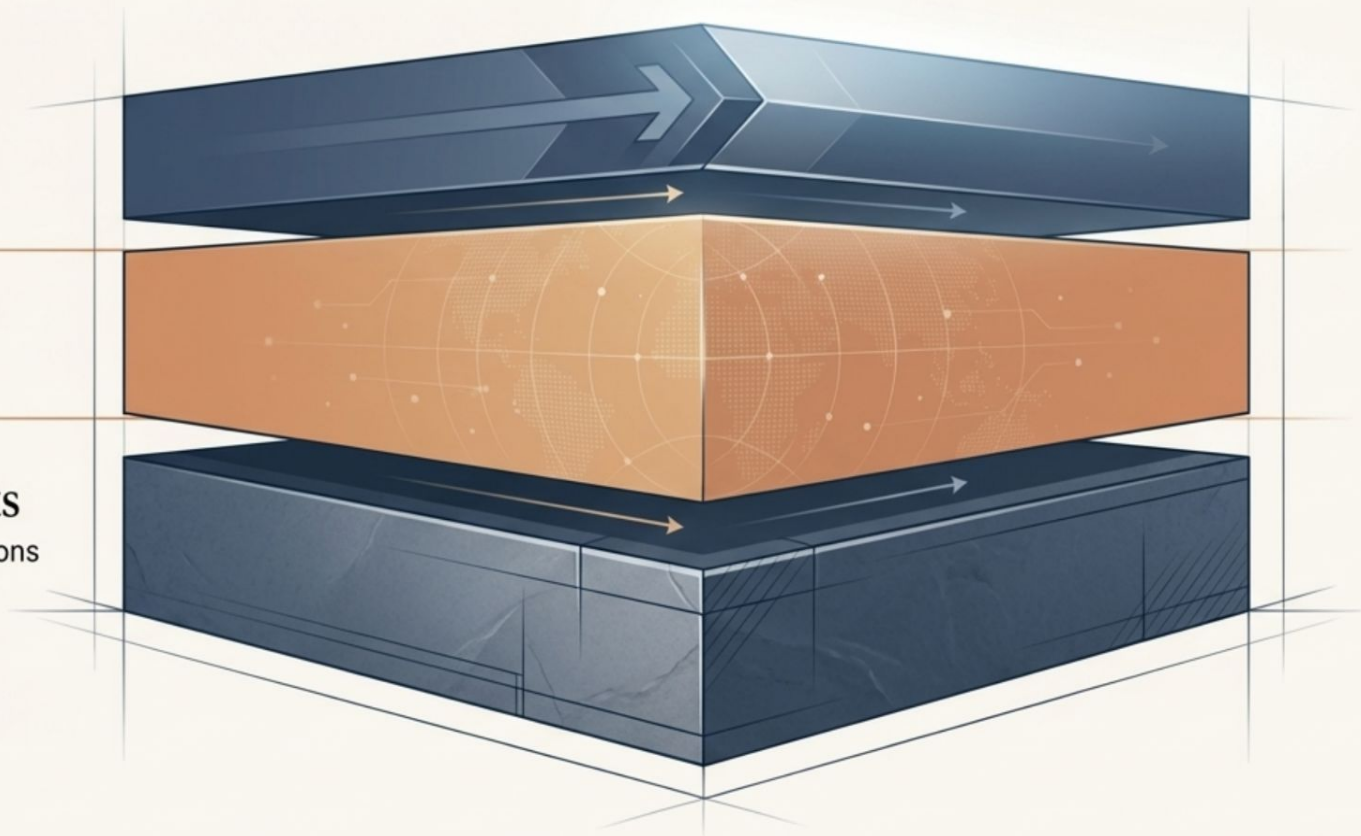
Emerging AI-Specific Regulations

The Universal Baseline

Data Privacy Regulations

Foundational Blueprints

Proven Industry-Specific Regulations



We're Not Starting from Scratch: Lessons from Finance & Pharma

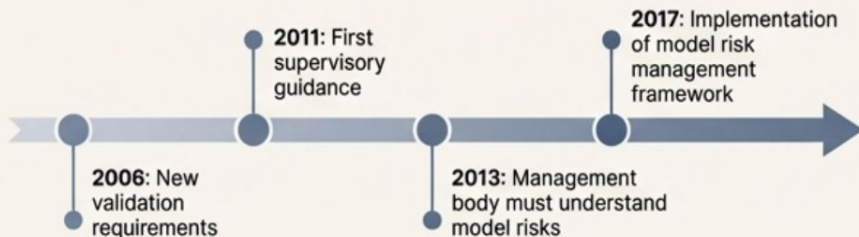
Highly regulated industries have practiced **rigorous model governance** for years, offering established principles we can adopt for MLOps.



Finance (Model Risk Management - MRM)

Driving Force: Mitigating systemic financial risk and preventing catastrophic losses.

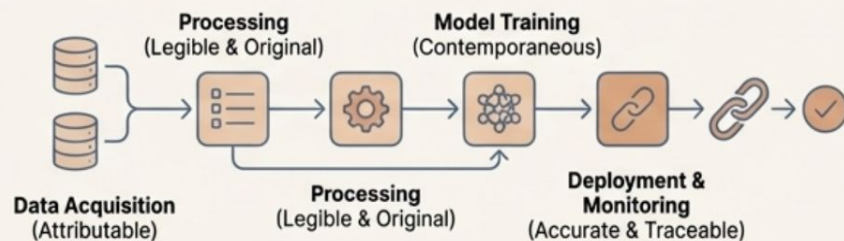
- Formal **Model Inventory**
- Established **Risk Governance Frameworks**
- Robust **Life Cycle Management**
- Mandatory **Independent Validation & Review**



Pharma (GxP - Good Practice guidelines)

Driving Force: Ensuring patient safety and product efficacy.

- Absolute **Traceability** (re-creating development history)
- Clear **Accountability** (who did what, when)
- Proven **Data Integrity** (ALCOA: Attributable, Legible, Contemporaneous, Original, and Accurate)



Data Privacy Laws Have Set a New Global Standard for All

If your models use personal data, GDPR and CCPA mandate a baseline of governance. The burden of proof is on the organization to demonstrate compliance.

Key MLOps Implications



Purpose Limitation & Data Minimization: You must justify the data you collect and use for modeling.



The Right to Explanation: A direct driver for explainability tools. Organizations must be able to explain automated decisions that have a significant effect on individuals.



Traceability for User Rights: You must track data lineage to fulfill user requests like the "right to be forgotten."

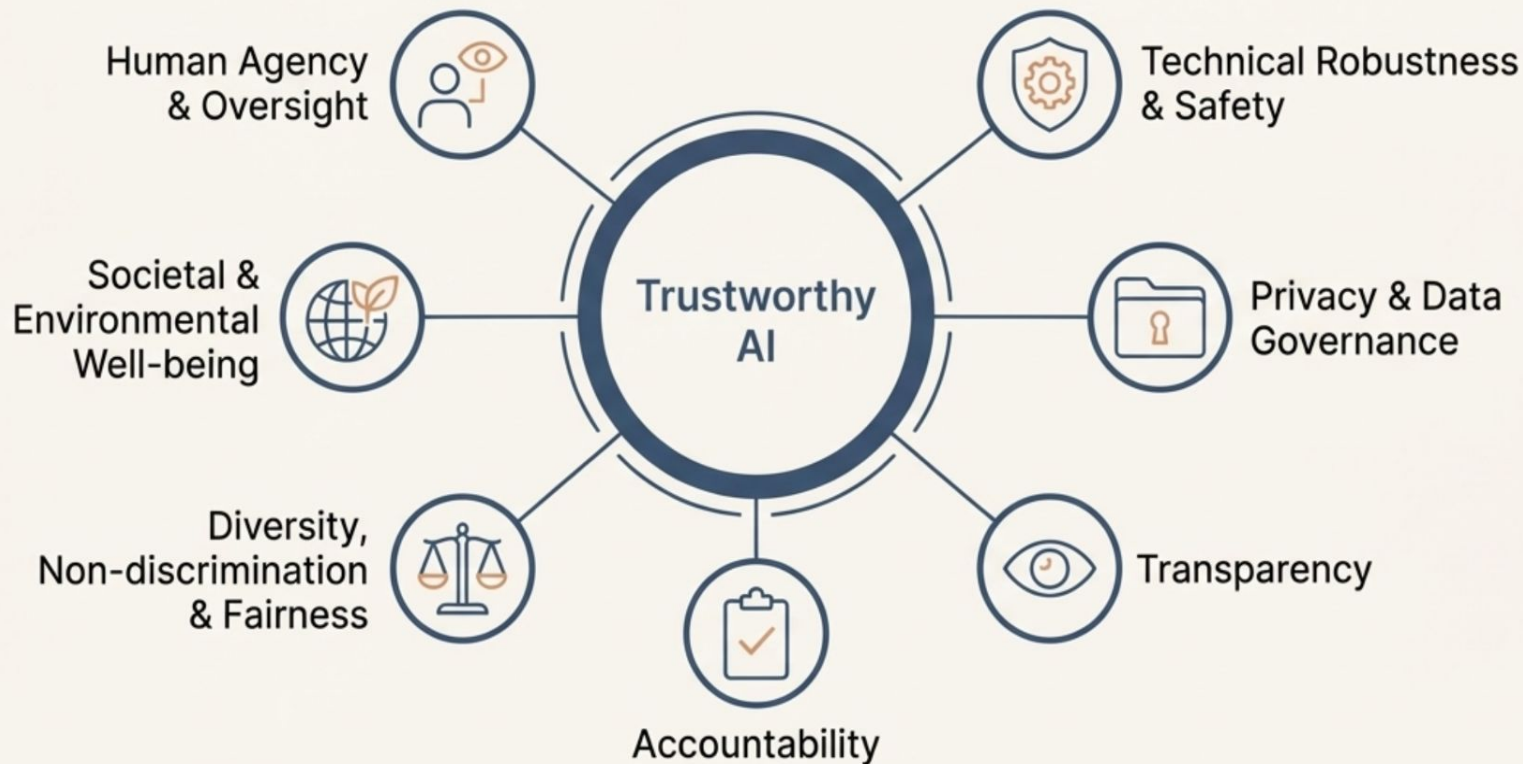


Accountability: Organizations must be able to centrally control, manage, and audit all data processing activities.

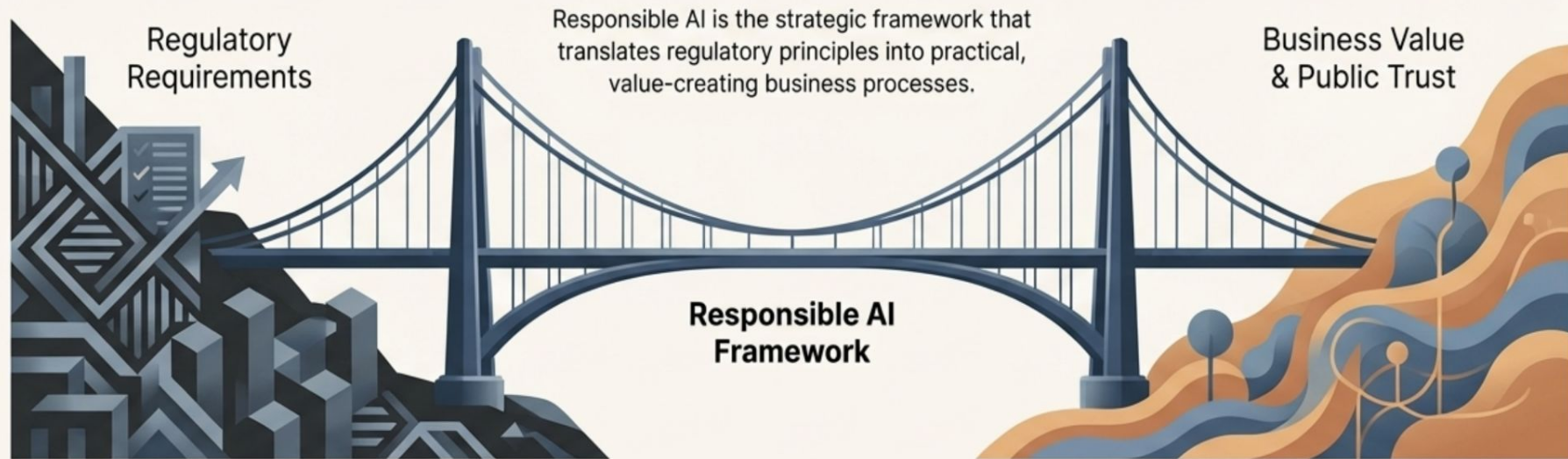


The Future is AI-Specific: Preparing for 'Trustworthy AI'

The next wave of regulation moves beyond just data to govern the AI systems themselves, with the EU's proposed framework setting the global benchmark.



Bridging the Gap from Regulation to Reality with Responsible AI



Definition:

A proactive approach to developing ML systems that are **Accountable**, **Sustainable**, and **Governable**.

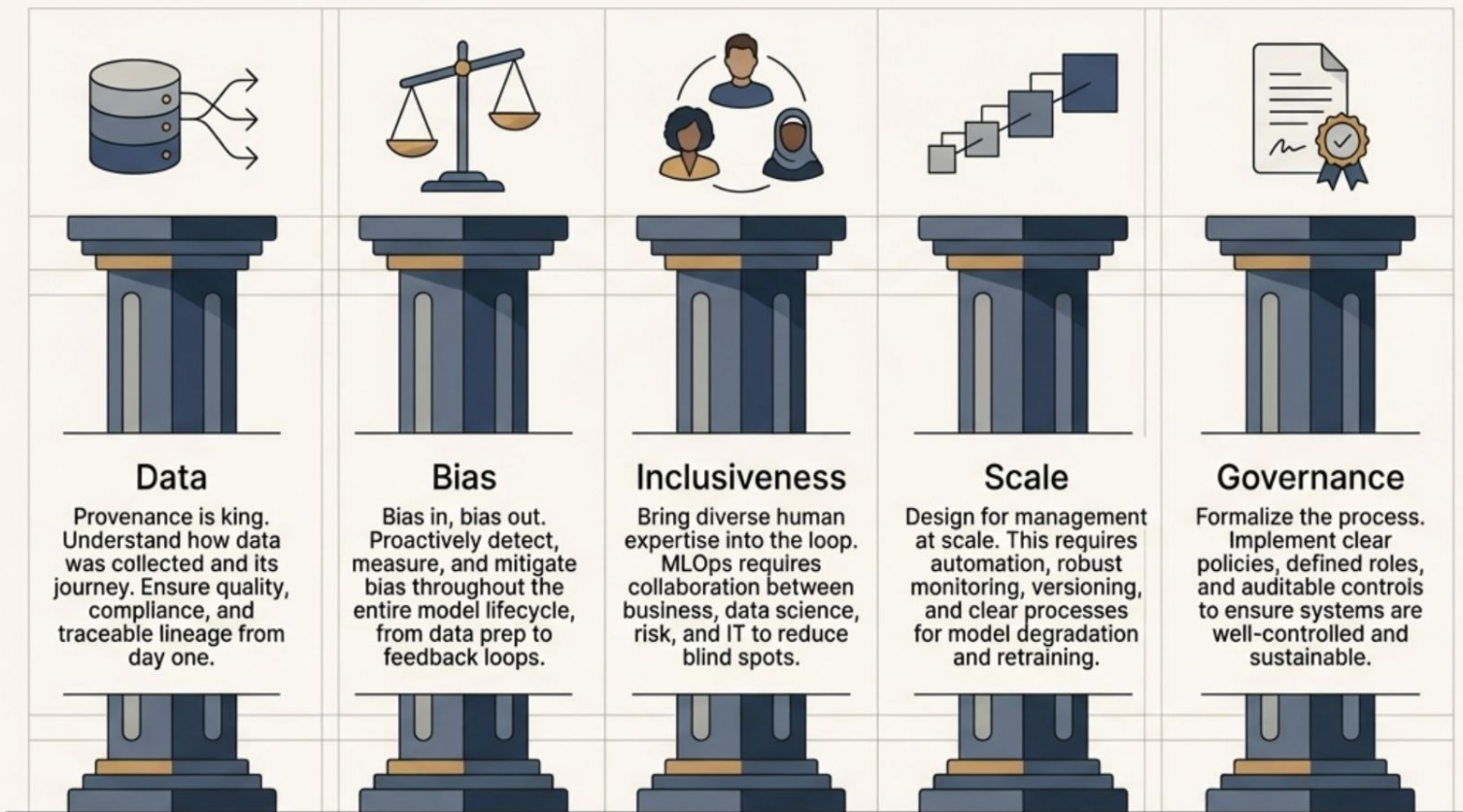


The Mindset Shift:

Moving from reactive compliance to a proactive strategy for building systems that are intentionally fair, transparent, and aligned with ethical principles.

The Five Pillars of a Responsible AI Strategy

These five elements are the foundation of a world-class MLOps governance program.

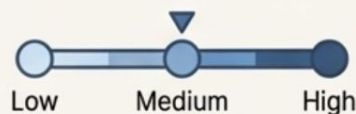


MLOps Governance Framework

Phase 1: Strategy & Design

1. Classify Use Cases

Assess the risk level of every project. Categorize initiatives to understand their unique governance needs, from ad-hoc governance needs, from ad-hoc analytics to mission-critical operational systems.



2. Establish an Ethical Position

Go beyond the law. Define and document your company's stance on fairness, transparency, and the potential societal impact of your AI.



3. Establish Responsibilities

Use a RACI matrix (Responsible, Accountable, Consulted, Informed) to clarify ownership for each stage of the MLOps lifecycle.

Tasks	Data Scientists	Risk/Audit	Business Stakeholders	Data Ops
Data Preparation	R	I		C
Tasks	R		A	C
Model Acceptance		A	C	I
Deployments		I	R	

R: Responsible, A: Accountable, C: Consulted, I: Informed

4. Determine Governance Policies

Create specific, documented rules for reproducibility, audit, testing, and sign-offs that align with the risk level of the use case.



MLOps Governance Framework

Phase 2: Implementation & Operations

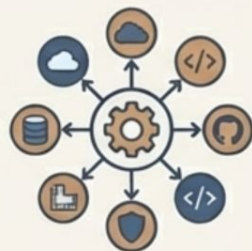
5. Integrate into the MLOps Process

Embed policy checks, quality gates, and human-in-the-loop sign-offs directly into your CI/CD pipelines and development workflows.



6. Select Centralized Tools

Use a single platform to manage, track, and enforce the governance process. This provides a central point of reference and a single source of truth for audits.



7. Engage and Educate

Governance is culture. Train everyone—from business analysts to data scientists to IT—on the policies and their specific role in upholding them.



8. Monitor and Refine

Governance is a living process. Continuously review its effectiveness and adapt to new risks, evolving public opinion, and emerging regulations.



GDPR Core Principles

The GDPR sets out specific principles for the processing (collection, storage, alteration, and use) of personal data, which have a direct impact on how ML models are trained and managed:

- **Lawfulness, fairness, and transparency:** Data must be processed legally and clearly.
- **Purpose limitation:** Data should only be used for the specific purpose for which it was collected.
- **Data minimization:** Only the data necessary for the specific task should be processed.
- **Accuracy:** Personal data must be kept accurate and up to date.
- **Storage limitation:** Data should not be kept longer than necessary.
- **Integrity and confidentiality (security):** Organizations must ensure the security of the data they process.
- **Accountability:** Organizations are responsible for demonstrating compliance with these principles.

ML-Specific Guidelines and Implications

The GDPR highlights several practical requirements for Data Science teams to remain compliant with these regulations:

- **Data Lineage Tracking:** Compliance requires understanding the **data lineage** at the feature level to ensure that no **personally identifiable information (PII)** is used inappropriately to train a model.
- **Anonymization Limits:** Simply anonymizing or pseudo-anonymizing data is not always a sufficient solution, as it may still be possible to single out individuals, which would violate GDPR requirements.
- **Bias and Discrimination:** A primary goal of these regulations is to limit **potential discrimination** against individuals through the misuse of personal data in ML systems.
- **Governance Frameworks:** Organizations must implement robust governance strategies to manage datasets used in ML models and ensure safety and accountability.

Digital Personal Data Protection (DPDP) Act, 2023

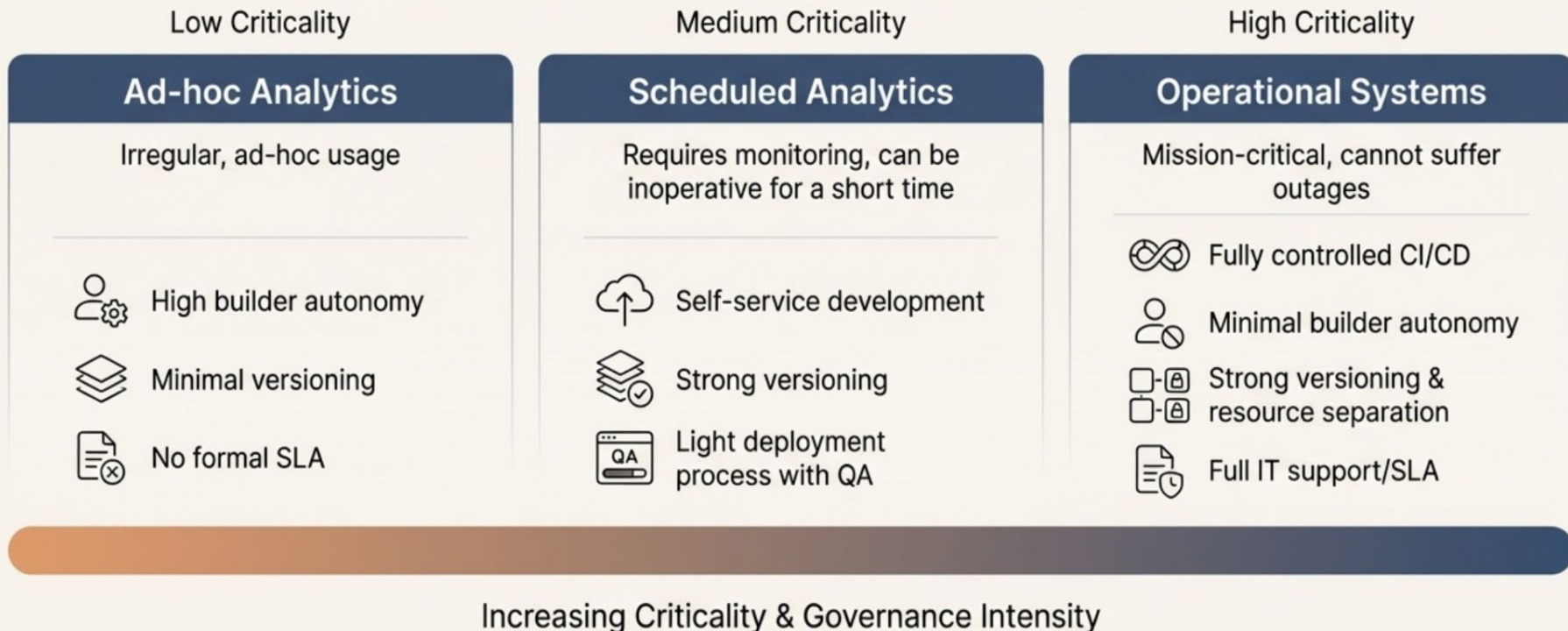
- **Applicability**
 - Applies to personal data collected digitally in India, or offshore if offering goods/services to India (excluding personal use or publicly available data).
- **Consent & Notice**
 - Data processing requires free, specific, and informed consent, preceded by a notice explaining the exact data and purpose.
- **Legitimate Uses (No Consent)**
 - Consent is not needed for specific situations like medical emergencies, employment purposes, or government services (subsidies/licenses).
- **Rights of Individuals (Data Principals)**
 - Users have the right to access their data, request correction/erasure, and seek grievance redressal.
- **Duties of Companies (Data Fiduciaries)**
 - Companies must implement security safeguards, ensure data accuracy, and delete data once the purpose is fulfilled.
- **Data Breach Notification**
 - In case of a breach, companies must strictly notify both the Data Protection Board and the affected users.
- **Significant Data Fiduciaries (SDFs)**
 - Large-scale entities (like Big Tech) face extra rules, such as appointing a Data Protection Officer and conducting impact assessments.
- **Protection of Children**
 - Verifiable parental consent is mandatory; tracking, behavioral monitoring, or targeted ads for children are strictly banned.
- **Penalties**
 - Violations attract fines up to ₹250 Crore (e.g., for security failures), but there are no criminal jail terms.

DPDP ACT - Relevance to ML, AI, and Deep Learning

- **Training Data Constraints**
 - You cannot use customer data for "model training" unless that specific purpose was explicitly declared in the consent notice.
- **Public Data Exemption**
 - Scraping data that users have voluntarily made "publicly available" (e.g., public tweets/profiles) is likely exempt from strict consent rules.
- **The "Unlearning" Challenge**
 - If a user exercises their "Right to Erasure," AI companies may legally need to retrain models or use "machine unlearning" to remove that user's influence from the neural network.
- **Automated Decision Making**
 - If an AI model (e.g., for loans/hiring) makes a decision affecting a user, the company is legally liable if the data used was inaccurate or incomplete.
- **Algorithmic Audits (DPIA)**
 - Significant Data Fiduciaries deploying high-risk AI must conduct Data Protection Impact Assessments to mitigate privacy risks before deployment.
- **Research Loophole**
 - AI researchers can process personal data without strict consent, provided the model is not used to make decisions about specific individuals (pure research).
- **Model Security**
 - Companies must secure not just the database but also the inference pipeline to prevent attacks (like Model Inversion) that could leak training data.

Governance is a Spectrum, Not a Switch

Apply your framework intelligently. The intensity of governance should be proportionate to the risk and criticality of the project.



Key Principles for Navigating the Future of MLOps



Governance is a Strategic Imperative

It is no longer optional. It's the foundation for managing business risk, building public trust, and complying with a multi-layered regulatory landscape.



The Future is 'Trustworthy AI'

Emerging AI-specific regulations are codifying principles of fairness, transparency, and accountability. Build these capabilities into your MLOps DNA now to stay ahead of the curve.



Adopt a Structured, Risk-Based Framework

Don't improvise. Implement a formal, centralized governance program based on Responsible AI principles. Match governance intensity to risk to enable scale and create a sustainable competitive advantage.

Thank you